# Kibo-RPC 7th — Team Timeline & To-dos

Nov 10, 2026 · @Timothy Tan

Our program must be submitted by Jan 20, 2027, 10:59 PM Malaysia time (23:59 JST). Plan: a safe baseline ready to submit by Dec 6, 2026, feature freeze on Jan 11, 2027, then only tuning.

## Key dates

| Date | What | Notes |
|---|---|---|
| 15 Oct 2026 | Web simulator opened | Already live — log in at jaxa.krpc.jp/user-auth |
| Mid–late Jan 2027 (TBA) | "Preliminary" simulator opens | Must run the final program here before pressing SUBMIT |
| 20 Jan 2027, 11:00 AM MYT | Last day to send questions to JAXA | Later questions get slow replies |
| 20 Jan 2027, 10:59 PM MYT | Program submission deadline | Only "FINISHED" runs can be submitted; resubmit freely until then |
| 21 Jan – 17 Feb 2027 | Malaysia preliminary round | Run by MYSA; score = average of 10 random runs |
| Early Mar – mid Apr 2027 | Finalists modify code for the ISS | Short, intense revision window |
| Early May 2027 | ISS final run on the real Int-Ball2 | One run, no restarts |
| June 2027 | Final Round Event | Results and rankings announced |

Open questions: when exactly the Preliminary simulator opens, and whether MYSA adds its own judging criteria (the rulebook says some countries do) — check mysa.gov.my/kibo-rpcmalaysia.

## Roles

Split five people into three workstreams, with one person also acting as team lead (the rulebook requires a leader who submits and talks to MYSA).

| Role | People | Owns |
|---|---|---|
| Lead + infra/testing | 1 | Repo, packaging, web and local simulator, sim log, submission, contact with MYSA |
| Movement + route | 2 | Moving between checkpoints, visit order, time budget, recovery when moves fail |
| Vision + alignment | 2 | ArUco marker detection, pose estimate, fine-tuning position and roll in front of each marker |

Team (5): @Timothy Tan, Hao Zhe, Nichole, Rachel, Yu Xin. Who takes which role is still to be decided.

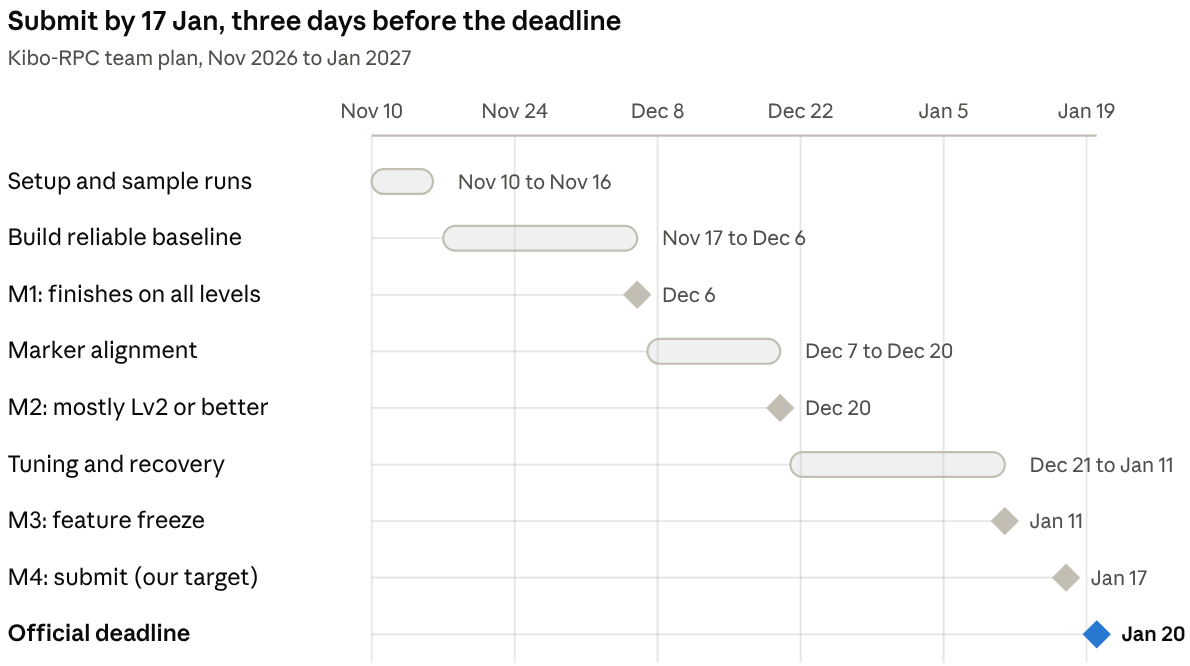
## Week-by-week plan

Ten weeks: first make it finish, then make it accurate, then make it fast and safe. Shift weeks around exams as needed.

| Week | Dates | Focus | Done when |
|---|---|---|---|
| 1 | 10–16 Nov | Setup: everyone runs the unchanged sample on the web simulator; push `users/` to GitHub with `.gitignore`; add CLAUDE.md and docs; confirm roles | All 5 people have a FINISHED sample run |
| 2 | 17–23 Nov | Read rulebook + Programming Manual §5–6. Movement: baseline that visits every checkpoint, reaches the goal and reports finished. Vision: save camera frames and detect marker IDs | Baseline finishes at least once |

| Week | Dates | Focus | Done when |
|---|---|---|---|
| 3 | 24–30 Nov | Movement: timeouts on every call; time guard that always reports before 5 min. Vision: pose (distance + angle) from `camera_info`. Infra: local simulator working | Baseline finishes 3 runs in a row |
| 4 | 1–6 Dec | Movement: shortest visit order. Everyone: log every run in sim-log.md | **M1: baseline finishes on every level** |
| 5 | 7–13 Dec | Vision: correct position, centering and roll using the marker until it hits the target level | Alignment works at one checkpoint |
| 6 | 14–20 Dec | Merge alignment into the main flow; check which level each checkpoint reaches | **M2: most checkpoints score Lv2 or better** |
| 7 | 21–27 Dec | Holiday week, light load: batch simulator runs, fix the most common failure | Top failure fixed |
| 8 | 28 Dec–3 Jan | Tune speed vs accuracy: the time multiplier (up to 2×) vs higher levels; decide how long to spend per checkpoint | Best average score so far |
| 9 | 4–11 Jan | Recovery: retry or skip a stuck checkpoint, handle drift from the microgravity disturbance | **M3: feature freeze on 11 Jan** |
| 10 | 12–20 Jan | Bug fixes and tuning only; many runs. Run the best version on the Preliminary simulator and SUBMIT; send last questions to JAXA | **M4: submitted by 17 Jan**, 3 days of buffer |

# Milestones




milestones · 10 Nov 2026 to 20 Jan 2027

Baseline locked by 6 Dec, features frozen on 11 Jan, and our own submit target of 17 Jan leaves three days of buffer before the official deadline.

# Every week

- One 30-minute team sync: last week's best score, blockers, this week's owners
- Log every simulator run in `docs/sim-log.md`: date, commit, level, score, what changed
- One branch per change, merged by pull request, so `main` always finishes
- Confirm a fix on the web simulator, not just the local one (the web one runs on ARM like the real robot)
- Update `docs/architecture.md` when the program's flow changes
- Never commit the simulator login

## After submission

If we make the finals, JAXA reviews our source code for safety and usually asks for changes, so keep early March to mid April clear. The ISS run in early May is a single run with no restarts, so robustness matters more than in the preliminary round.